

FACULTY OF SCIENCE AND TECHNOLOGY

LOGIC BUILDING PROGRAMMING & VERSION CONTROL REPORT

PROGRAMME: BSSE/BSCS/BIT

YEAR/SEM: YEAR 2 SEM 1

COURSE CODE: BIT2115

PROBLEM NAME: JAVA VARIABLE TYPES

STUDENT DETAILS:

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REG NUMBER	25/1512/BSCS-S

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Section 1

Logic Building & Programming

Lines of code & explanations

Line Number	Explanation
#1	Declares the Course class.
#2-3	Declares instance variables for the course name and registered students; each object has its own values.
#4	Declares the static totalStudents variable shared by all Course objects.
#6-10	Constructor initializes each course object and adds its registered students to the shared total.
#12	Declares displayStatistics(), which displays statistics for one course.
#13-14	Declares a local variable, attendancePercentage, and calculates the attendance percentage temporarily.
#16-18	Displays the individual course name, registered students, and attendance percentage.
#21-23	Declares a static method that displays the university-wide total.
#26-28	Creates three Course objects with different course names and registration numbers.
#30-32	Calls displayStatistics() for each course using the object reference.
#34	Calls the static method to display the total across all courses.

Detailed Screenshots and caption descriptions

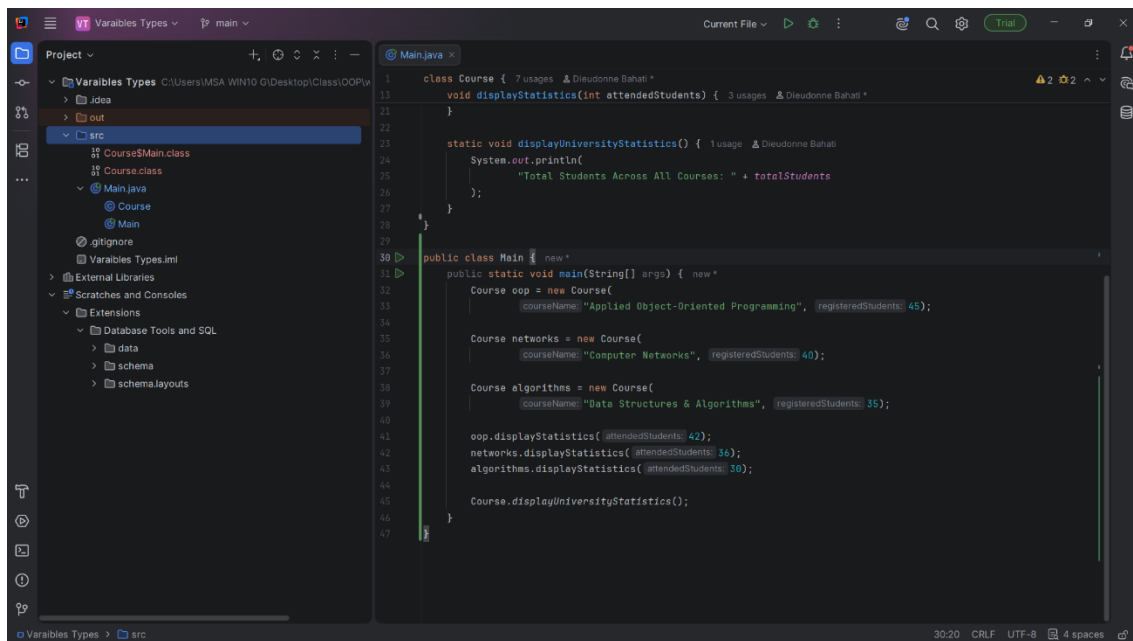
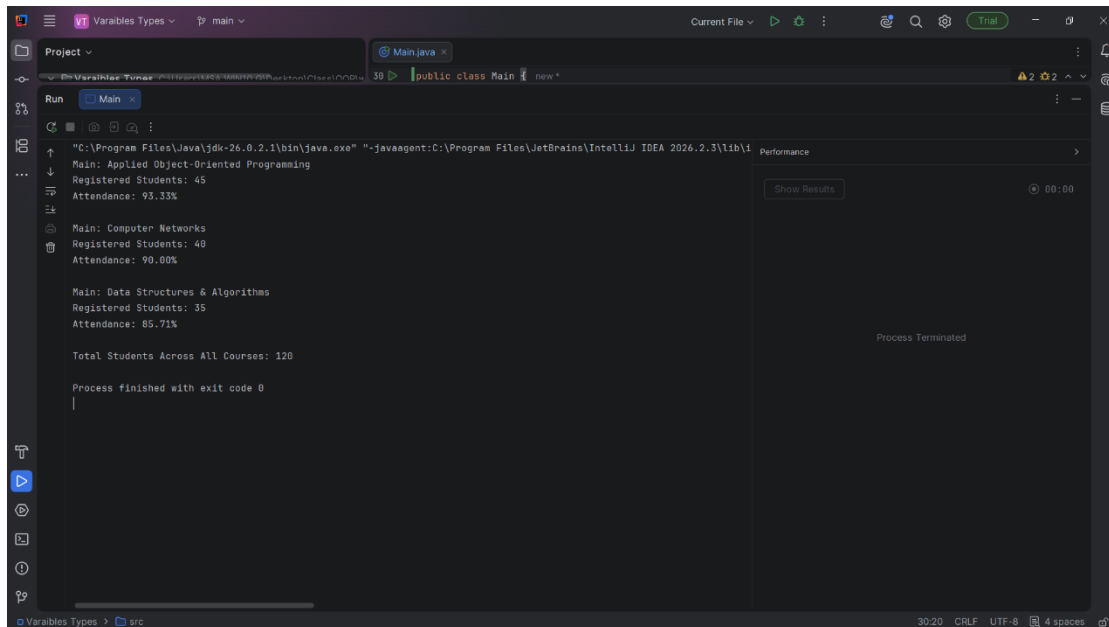


Figure 1:Class Main showing project structure

Output and caption descriptions



```
Run: C:\Program Files\Java\jdk-26.0.2.1\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2024.2.3\lib\i...
Main: Applied Object-Oriented Programming
Registered Students: 45
Attendance: 93.33%

Main: Computer Networks
Registered Students: 40
Attendance: 90.00%

Main: Data Structures & Algorithms
Registered Students: 35
Attendance: 85.71%

Total Students Across All Courses: 120

Process finished with exit code 0
```

Figure 2: Output after running the Java enrolment program

Output Explanation

The program creates three course objects: Applied Object-Oriented Programming with 45 students, Computer Networks with 40 students, and Data Structures & Algorithms with 35 students. The displayStatistics method uses a local attendancePercentage variable for a temporary calculation. The static totalStudents variable is shared by all Course objects, so the university-wide total is 120 students. Each course object keeps its own courseName and registeredStudents values.

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Section 2

Version Control

Repository Overview

Repository Name	OOP
Repository URL	C:\Users\MSA WIN10 G\Desktop\Class\OOP\work\Variables Types
Branches	main

Commit #1 On Branch #1

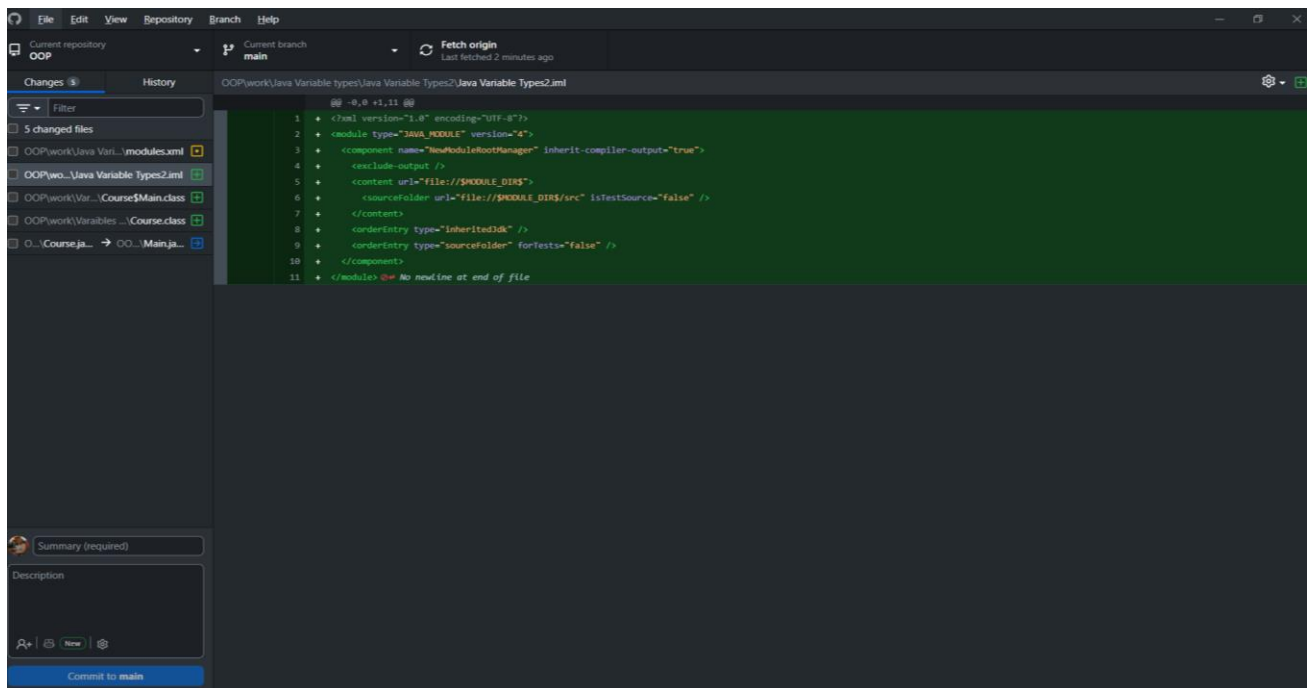


Figure 3: Initial commit on the main branch

Commit Summary

Commit Name	New: I created ne class Main instead of just course
Commit ID	c90da6f406660dbec91dc87a184f94cbe89cc9cd
Commit Description	New: I created ne class Main instead of just course
Branch Affected	main

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Section 3

What Have You Learned From The Task?

Skill Set	Description
Local Variables	Used inside a method for temporary calculations; attendancePercentage exists only while displayStatistics executes.
Instance Variables	Store data unique to each Course object, such as courseName and registeredStudents.
Static Variables	A single shared variable, totalStudents, keeps the total number of students across all Course objects.